

Question 2 (20 Marks)

2.1 Consider the following Java program:

```
01 //Program 1
02 public enum Color {
03     RED("FF0000"), GREEN("00FF00"), BLUE("0000FF");
04         0           1           2
05     private String hexCode;
06
07     private Color(String hexCode) {
08         this.hexCode = hexCode;
09     }
10
11     public String getHexCode() {
12         return hexCode;
13     }
14     public static void main(String [] a) {
15         for (Color color : Color.values())
16             System.out.println((color.ordinal() + 1) + " " + color +
17                               " has code " + color.getHexCode());
18     }
19 }
20
```

Based on the program above that defines an enumerated type called **Color**, answer the following questions:

a) What is the **significance** of the values "FF0000", "00FF00", and "0000FF" that are passed as arguments to the constructor of each value in the **Color** enumerated type?

- hexadecimal color codes that represent the color values for RED, GREEN and BLUE respectively (1 mark)

b) What is the **purpose** of the **getHexCode()** method in the **Color** enumerated type?

- accessor that returns the hexadecimal color code (String) (1 mark)

c) What will be the output of the program?

(3 marks)

d) Write a Java statement in **main()** method that uses the **compareTo()** method to display the results of a comparison of the positions of **RED** and **BLUE** in the **Color** enumerated type. What will the output be?

(2 marks)

(c) 1) RED has code FF0000
2) GREEN has code 00FF00
3) BLUE has code 0000FF

(d) $\text{System.out.println}(\text{Color.RED.compareTo}(\text{Color.BLUE}));$

Output : -2

$= \text{RED.ordinal}() - \text{BLUE.ordinal}()$

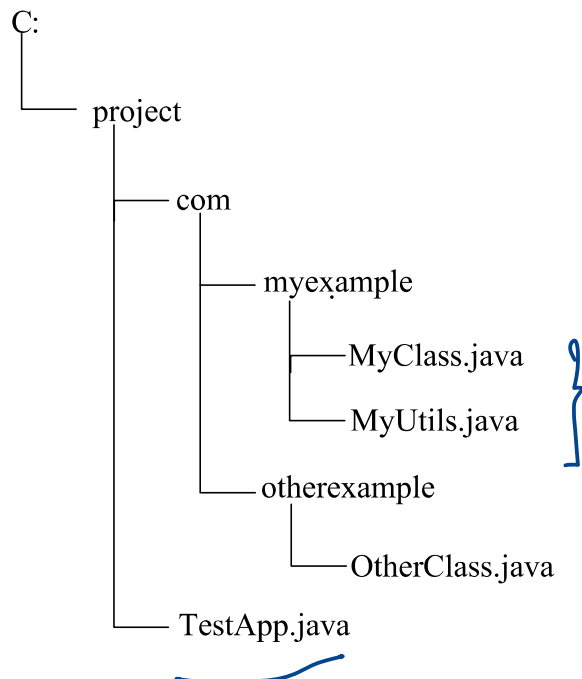
$= 0 - 2 = -2$

e) Consider adding the following statements in the `main()` method:

```
if (!Color.GREEN.equals(Color.RED))  
    System.out.println("GREEN and RED are two different colors.");  
else  
    System.out.println("GREEN and RED are the same color.");
```

The statements, however, include an error. What is the error, and how can it be fixed?
(2 marks)

2.2 Given the directory structure as shown below:



a) What is the package name of `MyClass.java` and `MyUtils.java`?

`com, myexample`

(1 mark)

b) What is the package name of `OtherClass.java`?

`com, otherexample`

(1 mark)

c) In `TestApp.java`, how would you import all the classes in `com.myexample`?

`import com.myexample.*`

(1 mark)

d) Suppose you want to move `MyUtils.java` to a new package called `com.myexample.util`. What changes would you need to make to the directory structure and the code?

(3 marks)

1) Create a new subfolder : `project/com/myexample/util`

2) Package declaration in `MyUtils.java` : `package com.myexample.util;`

3) Update import statements in `TestApp.java` : `import com.myexample.util.
MyUtils;`

2.3 Consider the Java program below and answer the following questions:

```
01 //Program 2
02 public class WrapperClass {
03     public static void main(String[] args) {
04         Integer num1 = 10;
05         Double num2 = 3.14;
06
07         //(a)Converting an object of a wrapper type to
08         //its corresponding primitive value
09         int value1 = num1.intValue();
10         double value2 = num2.doubleValue();
11
12         System.out.println("value1: " + value1);
13         System.out.println("value2: " + value2);
14
15         //(b)Adding value1 and value2 and storing the
16         //result in the result variable by converting
17         //it to the Double wrapper class
18         Double result = Double.valueOf(value1 + value2);
19
20         System.out.println("result: " + result);
21     }
22 }
```

- a) Fill in the blanks with an appropriate Java statement to convert an object of a wrapper type to its corresponding primitive value. **Note:** you must use a method provided in the wrapper class. What is the name of the conversion?

(3 marks)

- b) Fill in the blanks with an appropriate Java statement to add **value1** and **value2** and store the result in the **result** variable by converting it to the **Double** wrapper class. **Note:** you must use a method provided in the wrapper class.

(2 marks)